INCH-POUND

MIL-PRF-32020(USAF)
1 MARCH 1998

PERFORMANCE SPECIFICATION

TESTER, LIQUID VOLUMETRIC POSITIVE DISPLACEMENT METER,

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The tester is used to verify readings on displacement meters such as permanently installed pit meters in the Panero hydrant systems, permanently installed meters used with truck fill stand installations, or meters mounted on hydrant refueling equipment, such as the MH-2 and MH-3 fuel servicing trailers used in conjunction with the Pritchard type hydrant systems. The tester is designed for the verification of positive displacement meters within the operational flow range of 30 gpm to 600 gpm at operating pressures not exceeding 150 psi. The meter tester consists of a trailer chassis and running gear that contains a master meter, suitable inlet and outlet piping and connections, discharge hoses and nozzles, control valves and pressure and temperature gages. It can also be used to service aircraft at locations where totalizers are not available.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standard (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, 485 Quentin Roosevelt Road, Kelly AFB, TX 78241-6424 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6680

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SPECIFICATIONS

FEDERAL

A-A-50696	Reels, Static Discharge Grounding, 50 and 75 feet Cable Lengths
A-A-52030	Nozzles, Fuel and Oil Servicing, Non-automatic Shutoff and Nozzles, Fuel Servicing, Automatic Shutoff
GG-M-2742	Meter, Volumetric, Positive Displacement, Liquid, Aircraft Fuel, 600 GPM

DEPARTMENT OF DEFENSE

MIL-PRF-5624	Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST
MIL-PRF-85285	Coating, Polyurethane, High Solids
MIL-H-6615	Hose Assembles, Rubber, fuel and Water, with Reattachable Couplings, Low Temperature
MIL-C-46722	Chassis, Trailer: 1 to 5 Ton, 4 Wheel, General Purpose

STANDARDS

FEDERAL

FED-STD-595	Colors Used In Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MS25384	Electrostatic Discharger Jumper, Fuel Nozzle-to-Aircraft

(Unless otherwise indicated, copies of Federal and Military Specifications, Standards, and Handbooks are available from the Defense Printing Service Detachment Office, Building 4D, NPM-DoDSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

DRAWING

Air Force, 54J6207	Trailer Chassis, 2000-Pound Capacity, 4-Wheel, Type MB-1

(Copies of this is available from SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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American Society for Testing and Materials (ASTM)

F1347 Manually Operated Fueling Hose Reels, Standard Specification for

(Application for copies should be addressed to the American Society of Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA. 19428-2959.)

Society of Automotive Engineers

SAE-AS8090 Mobility, Towed Aerospace Ground Support Equipment, General Requirements for

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Dr. Warrendale, PA 15096-0001

2.4 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection.

3.2 Nozzle. The nozzle connected to the hose on the hose reel shall meet or exceed the requirements of A-A-52030, Type I, size 3, class A, style 2.

3.2.1 Electrostatic discharge jumper. The nozzle shall include a MS 25384-1 electrostatic discharge jumper.

3.3 Hose reel. The hose reel shall meet or exceed the requirements of F1347, Type 4 and shall contain minimum 50 feet of 2 ½ inch of hose that conforms to MIL-H-6615.

3.4 Meter assembly. The volumetric positive displacement meter assembly shall be in accordance with GG-M-2742. The meter shall include an automatic pre-set stop feature that will allow the operator to predetermine an amount of fuel to register and then automatically shut off the flow without causing a shock or hammer to the plumbing . The meter shall incorporate an emergency shut-off system.

3.5 Single-point hose assembly. A single point hose assembly shall be provided that is connected to the swivel joint and that is 15 feet long and has an inside diameter of 3 inches. Male couplings at each end shall conform to MIL-H-6615. A stowage facility shall be provided to coil the hose when not is use.

3.6 Swivel joint. A swivel joint shall serve as a connection between the single point hose assembly and the trailer flow system.

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3.7 Pressure fuel servicing adapter. The adapter shall connect the meter assembly to the inlet. It shall have a 4-inch plug mounted a right angle to the adapter and a dust cap to prevent the entrance of foreign matter when the trailer is not in use.

3.8 Instrumentation. Instrumentation to monitor temperature and pressure through the operating range of the tester shall be provided and must be readable from a distance of 10 feet in average light conditions.

3.9 Static Grounding Reel. The reel shall be used to provide a contact between the trailer and the ground. The assembly shall be in accordance with specification A-A-50696, Type II.

3.10 Valves. Valves of the appropriate type shall be used to close off the single point and hose reel delivery system when necessary.

3.11 Chassis and running gear. Chassis and running gear shall conform to MIL-C-46722 and Air Force drawing 54J6207.

3.12 Parking Brake. Parking brakes shall perform to SAE-AS8090.

3.13 Weight. The tester shall not exceed 1200 lbs. Gross dry weight.

3.14 Appearance. The unit and all components shall be painted in a tone down color of dark green, Federal Standard 595, Color No. 24052, polyurethane coating MIL-PRF-85285.

3.15 Workmanship. The tester shall be manufactured meeting or exceeding standard commercial practices for the selection and installation of hardware, corrosion prevention practices and the ease of repair and servicing..

3.16 Performance. The tester shall be capable of withstanding exposure of the following conditions without adversely affecting the integrity, accuracy, and not cause any leakage.

a. Temperature ranges of -65° to +160°F.

b. Fungus growth as encountered in tropical climates.

c. A relative humidity of 95 ±5% at a temperature of +160°F.

d. Sand and dust particles as may be encountered in desert areas.

e. Atmosphere containing salt-laden moisture.

f. Being towed over semi-rough terrain and/or improved roads.

3.17 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

4 VERIFICATION

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4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see para. 4.2)

b. Conformance inspection (see para. 4.3)

4.2 First article inspection. First article inspection, when required, shall be performed on one complete tester and shall consist of the following 4.4 examination, 4.5 methods of inspection, and 4.6 performance with a repeat of 4.6.1 at the completion of tests described by 4.6.3 through 4.6.8..

4.3 Conformance tests. Each tester shall be subjected to the following tests as described under 4.4., 4.5.2 and 4.6.1.

4.4 Examination. Each tester shall be examined for compliance with the requirements specified in 3.1 through 3.15.

4.5 Methods of inspection.

4.5.1 Testing apparatus. Apparatus used in conjunction with the testing specified herein shall be laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.5.2 Fuel. Unless otherwise specified, the fuel used during the testing specified shall conform to MIL-F-5624 and shall have a specific gravity of not less than 0.750 at 60°F.

4.6 Performance.

4.6.1 Hydrostatic test. The tester shall be filled and subjected to a pressure of 225 psi for a period of 10 minutes. All leaks encountered shall be repaired and the unit retested. No leakage shall be permitted.
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4.6.2 Cross-country operation. The tester shall be towed at a speed of 15 mph and a distance of 1 mile over open fields, broken ground, loose sand or gravel, or uneven terrain. No meter inaccuracies, misalignment of components, or other detrimental effects shall result from this test.

4.6.3 Functional flow test. A test specimen meter shall be installed upstream of the tester in a pumping system capable of pumping at a rate of 600 gpm. Fuel shall be pumped through both meters at 6 flow rates from 100 gpm to 600 gpm at 100 gpm increments. Each tested flow rate shall last for a period of 1/2 hour. The pre-set stop will be set at 5000 gal and 10000 gal incrementally at a 600 gpm rate. During this test the meter on the tester shall function in a satisfactory manner. Two emergency stops shall be observed at the minimum and maximum flow rates.

4.6.4 High temperature test. The tester shall be subjected to high temperature in accordance with MIL-Std-810, method 501.1, procedure I, steps 1 through 4. Following the 48 hour exposure at 160° F. the tester shall be removed from the chamber and visually examined for deterioration. The

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tester shall show no deleterious effects from the exposure and shall complete the remainder of the tests in a satisfactory manner.

4.6.5 Low temperature test. The tester shall be subjected to low temperature in accordance with MIL-Std-810, method 502.1, procedure I, steps 1 through 6. Following the period of 24 hours after stabilization at -65° F. the tester shall show no evidence of failure and shall complete the remaining tests in a satisfactory manner.

4.6.6 Humidity test. The tester shall be subjected to humidity in accordance with MIL-STD-810, method 507.1, procedure I, steps 1 through 7. Following this test, the meter shall be visually examined for deterioration, evidence of swelling separation of components, and related types of failure.

4.6.7 Salt fog test. The tester shall be exposed to salt fog in accordance with MIL-STD-810, method 509.1, procedure I. During this test, the inlet and outlet of the meter shall be plugged to prevent internal contamination. The meter shall resist the effects of the test and shall show no evidence of failure or malfunction.

4.6.8 Sand and dust test. The tester shall be exposed to sand and dust in accordance with MIL-STD-810, method 510.1, procedure I, steps 1 through 4. During this test, the inlet and outlet of the meter shall be plugged to prevent internal contamination. Upon completion of the test, it shall be demonstrated that the tester is completely operable.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging shall be as specified in the contract or order (see 6.3). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Military-Unique Rational. There is no commercially available item to verify total fuel delivery suitable for single point and auxiliary tank refueling of military aircraft style systems. It must be capable of performing in the hostile environments of arctic and desert temperatures as well as maintaining its integrity on flightline surfaces as well unimproved surfaces of forward operating locations. It can also be used in servicing aircraft where local system totalizers are not in service.

6.2 Intended use. The type tester covered by this specification is intended to prove and verify readings of positive displacement meters installed in fixed hydrant refueling system and truck fillstands.

6.3 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number and date of this specification.
b. Applicable levels of preservation, packaging, and packing required.
c. Data specified herein should be submitted in accordance with DD-Form 1423.

6.4 Subject term (key word) listing.

Fuel
Totalizer
Trailer
Servicing
Flightline

Custodian:
Air Force-99

Preparing Activity:
Air Force-82

Agent Activity
Air Force -99

(Project 6680-F254)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT MIL-PFR-32020	2. DOCUMENT DATE 1 MARCH 1998

3. DOCUMENT
Tester, Liquid Volumetric Positive Displacement Meter

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	b. TELEPHONE *(Include Area Code)* (1) (210) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 Quentin Roosevelt Rd. Kelly AFB, TX 78241-6424	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS,** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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